

- a) What economic interpretation can be given to the vector $100b$?
- b) Suppose the company wishes to manufacture x_1 dollars worth of product B and x_2 dollars worth of product C. Give a vector that describes the various costs the company will have (for materials, labor and overhead).

Solution: a) Compute $100b = 100 \begin{bmatrix} 45 \\ 25 \\ 15 \end{bmatrix} = \begin{bmatrix} 4500 \\ 2500 \\ 1500 \end{bmatrix}$

The vector $100b$ lists the various costs for producing \$100 worth of product B — namely, \$45 for materials, \$25 for labor, and \$15 for overhead.

- b) The costs of manufacturing x_1 dollars worth of B are given by the vector x_1b , and the costs of manufacturing x_2 dollars worth of C are given by x_2c . Hence, the total costs for both products are given by the vector $x_1b + x_2c$. $\square$

Practice Problems

1. Prove that $u+v = v+u$ for any u and v in $\mathbb{R}^n$.

Solution: Let each element of u be represented by a_i , where $1 \leq i \leq n$.
Let each element of v be represented by b_i , where $1 \leq i \leq n$.

- Each element of $(u+v)$ will take the form $(a_i + b_i)$, where $1 \leq i \leq n$.
 - Each element of $(v+u)$ will take the form $(b_i + a_i)$, where $1 \leq i \leq n$.
 - Each element of $(u+v)$ will take the form $(a_i + b_i)$, where $1 \leq i \leq n$.
- $\therefore u+v = v+u$ (Commutative law for real numbers)

$\therefore u+v = v+u$ for any u, v in $\mathbb{R}^n$ $\square$

2. For what values of h will y be in $\text{Span}\{v_1, v_2, v_3\}$ if

$v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}$, $v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$, and $y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$